

Muhammad Mustafa

Senior Data Engineer

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Professional Summary

Senior Data Engineer with 9 years of experience designing and delivering scalable data pipelines, cloud platforms, and modern data warehouses. Expert in Python, SQL, Apache Spark, and ETL frameworks, building batch and real-time workflows on AWS, Snowflake, and Redshift. Skilled at transforming complex datasets into high-quality analytics-ready data, optimizing performance, ensuring reliability, and enabling actionable business insights. Proven leader mentoring engineers, standardizing processes, and improving platform scalability and maintainability.

Technical Skills

Programming & Scripting

Python SQL Scala Java Bash/Shell

ETL & Data Engineering

Data Pipelines ETL/ELT Data Transformation
Batch & Real-Time Processing Pipeline
Optimization Data Integration Data Governance

Cloud Platforms

AWS (S3, Glue, Redshift, Lambda, EMR, Athena, Kinesis, IAM, CloudWatch) Azure (Data Factory, Databricks, Synapse, Data Lake) GCP (BigQuery, Dataflow, Pub/Sub, Dataproc)

Databases

PostgreSQL MySQL SQL Server Oracle
MongoDB DynamoDB

Visualization & Analytics

Tableau Power BI Looker

Big Data & Processing

Apache Spark PySpark Hadoop Hive Kafka
Flink Spark Streaming

Workflow Orchestration

Apache Airflow Prefect Luigi AWS Step Functions

Data Warehousing & Modeling

Snowflake Redshift BigQuery Databricks
Lakehouse Delta Lake Star & Snowflake
Schema Metadata Management

DevOps & Tools

Git GitHub/GitLab Jenkins CI/CD Docker
Kubernetes Logging & Monitoring

Development Practices

Agile Scrum Code Review Documentation
Best Practices Implementation

Professional Experience

Senior Data Platform Engineer, ILSA Interactive

03/2022 – Present

- Designed, developed, and maintained scalable ETL pipelines using Python and SQL for multi-domain enterprise datasets, improving data reliability and accessibility.
- Built distributed Spark workflows for efficient transformation and aggregation, reducing processing times by 34% and supporting real-time analytics.

- Automated ingestion pipelines from APIs, databases, and external sources, increasing data freshness and consistency across platforms.
- Implemented Airflow orchestration with logging and alerting, proactively identifying pipeline failures and reducing downtime.
- Developed dimensional warehouse tables and star schemas, enabling fast, accurate, and consistent queries for BI teams.
- Migrated ETL workflows to AWS cloud services (S3, Glue, Redshift, Lambda), improving scalability, maintainability, and operational reliability.
- Created reusable Python modules for transformations, validations, and monitoring, enhancing development efficiency.
- Mentored junior engineers in ETL design, pipeline optimization, and cloud architecture best practices.
- Documented data pipeline architecture, transformation logic, and models for onboarding and internal governance.

Data Integration Engineer, Colabs

07/2020 – 02/2022

- Developed and optimized ETL pipelines using Python, SQL, and Spark, centralizing operational and transactional data for analytics and reporting.
- Built distributed Spark workflows to transform large-scale datasets for dashboards, BI, and executive reporting.
- Automated ingestion from APIs, relational systems, flat files, and streaming sources, improving timeliness and reliability of data.
- Designed validation frameworks and quality checks to monitor data accuracy, consistency, and completeness.
- Migrated legacy ETL pipelines to Spark, increasing performance and scalability while reducing processing times by 27%.
- Collaborated with analysts and product teams to translate reporting requirements into scalable, high-quality data models.
- Coordinated with DevOps teams to deploy pipelines and ensure stable production operations.

Analytical Data Systems Engineer, Fourspan Technologies

01/2018 – 06/2020

- Built ETL pipelines using Python and SQL to extract, transform, and load enterprise application data for reporting and analytics.
- Designed relational and analytical data models, star schemas, and warehouse structures to support structured reporting and BI workflows.
- Maintained batch processing pipelines and transformation scripts, ensuring data quality and schedule adherence.
- Implemented monitoring, error handling, and alert systems, improving pipeline reliability and minimizing downtime.
- Optimized ETL queries and transformations, reducing batch processing times by 23% and improving resource utilization.
- Conducted code reviews and documented workflows for team knowledge sharing and auditing purposes.
- Collaborated with analysts and senior engineers to refine datasets for improved usability and strategic reporting.

Education

Bachelor of Science in Computer Science